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Physicochemical Properties and Empirical Polarity Parameters of Lithium Salt-Based Deep Eutectic Solvents

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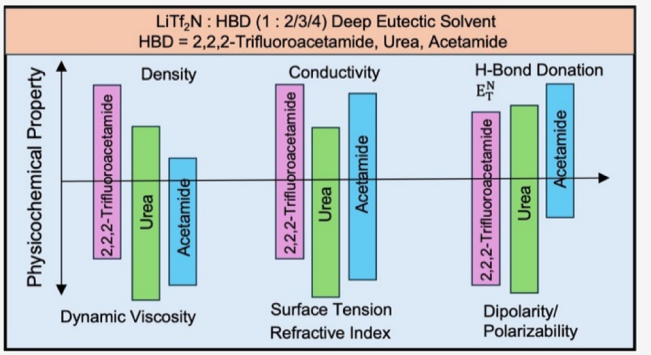
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ABSTRACT: Physicochemical properties of deep eutectic solvents (DESs) constituted of lithium *bis*(trifluoromethylsulfonyl)imide ($LiTf_2N$) as the H-bond acceptor (HBA) and three different H-bond donors (HBDs)—urea, acetamide, and 2,2,2-trifluoroacetamide—in (1:2), (1:3), and (1:4) mole ratios are investigated. The density of these DESs decreases linearly with increasing temperature within the 288−358 K range; it is higher for $LiTf_2N$/2,2,2-trifluoroacetamide in comparison to $LiTf_2N$/urea and $LiTf_2N$/acetamide. Density decreases linearly with increasing HBD per mole of $LiTf_2N$. The dynamic viscosity is highest for $LiTf_2N$/urea and lowest for $LiTf_2N$/2,2,2-trifluoroacetamide; it follows Vogel−Fulcher−Tammann (VFT) temperature dependence. With increasing HBD concentration, the dynamic viscosity of the DES decreases monotonically. Electrical conductivity, surface tension, and refractive index of these DESs at 298 K exhibit considerable variation with constituent and composition. The empirical solvent polarity parameter, E_T^N for all DESs within the investigated temperature range is close to or higher than one, implying these DESs to possess considerably high dipolarity/polarizability and/or HBD acidity. Kamlet−Taft empirical parameters—dipolarity/polarizability (π^*), HBD acidity (α), and HBD basicity (β)—reveal that the $LiTf_2N$/urea DES exhibits the highest π^*, followed by $LiTf_2N$/2,2,2-trifluoroacetamide and $LiTf_2N$/acetamide; the α is considerably higher for acetamide DES in comparison. The β values of these DESs are fairly low.



INTRODUCTION

Over the last two decades, deep eutectic solvents (DESs) have found their place as potent green alternatives to hazardous organic solvents.[1−4] Use of inexpensive and easily available materials has made them a popular choice for synthesis and purification purposes.[5,6] Till date, many DESs synthesized mainly from nature-derived substances are in occurrence. The more commonly used are those obtained by mixing quaternary ammonium salts as H-bond acceptors (HBAs) with several H-bond donors (HBDs), such as choline chloride:urea (1:2), choline chloride:glycerol (1:2), and choline chloride:ethylene glycol (1:2).[1,2] They have attracted immense popularity owing to their applicability in lithium-ion batteries (LIBs), CO_2 capture, metal extraction, synthesis of pharmaceutical compounds, catalysts in organic synthesis, and many more.[4,7−13] However, a relatively less explored family of DESs is the one formed by mixing lithium salts, such as lithium *bis*(trifluoromethylsulfonyl)imide ($LiTf_2N$), lithium hexafluorophosphate ($LiPF_6$), lithium nitrate ($LiNO_3$), or lithium chloride (LiCl), with urea- and acetamide-derived compounds.[14−17] $LiTf_2N$-based DESs are particularly more interesting due to their wide liquid-phase range and the ability of Tf_2N^- anion to lower the melting point of the salts.[18] Recently, a surge in the development and identification of the properties of these DESs has been noticed.

The first liquid binary combination of $LiTf_2N$ and urea was reported by Huang et al. in 2001 as a replacement for room-temperature molten salts (RTMS) consisting of alkyl-substituted nitrogen-containing cations, whose synthesis is neither easy nor cost-effective.[19] It is established that the higher affinity of Li^+ for the oxygen atom of the amide group weakens the interaction between Li^+ and Tf_2N^-, which further coordinates with the NH_2 group via H-bonding.[20] The presence of the CF_3 group in TF_2N^- renders the H-bonds even stronger. $LiTf_2N$, in a suitable combination with

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acetamide, is found to have a lower eutectic temperature than urea-containing DESs.[15,18] Physicochemical properties can be tuned by replacing acetamide with 2,2,2-trifluoroacetamide, which results in a change in H-bonding strength. Le and coworkers have suggested that the formation of DES in the $LiTf_2N$/2,2,2-trifluoroacetamide mixture is accompanied by the formation of a five-membered ring complex between Li^+, the O atom of the CO group, and the F atoms of the CF_3 group, since both O and F atoms are negatively charged.[20]

The fact that Li-salt-based DESs only contain Li cations is an intriguing characteristic. It is due to this reason, Smiatek et al. noted that, compared to other ionic liquid electrolytes, $LiTf_2N$ and urea-containing DESs may have a greater Li ion transport number.[14,21] Electrochemical studies in Li-salt-based DESs have shown promising outcomes. Conventionally used electrolytes in LIBs are mixtures of lithium salts (such as $LiPF_6$) with ethylene carbonates and ethyl methyl carbonate, which usually degrade and oxidize at higher voltages. In recent times, Li-salt-added choline chloride- based DESs have been gaining popularity in electrochemical applications, but the presence of bulkier choline ions prevents Li ions from transporting to the electrodes. Li-salt-based DESs have an advantage, as they do not contain any other cation. These and many other shortcomings can be overcome by the use of a suitably chosen DES as electrolyte. Li-salt-based DESs offer high thermal stability, ionic conductivity, and nontoxicity, along with cost-effectiveness, a large electrochemical window, and non-flammability.[22−25]

The Li-salt-based DESs are one step closer in the direction of a novel approach toward environmentally benign solvent systems. Since the discovery of Li-salt-based DESs, their potential to replace other solvent media, such as choline chloride-based DESs and ionic liquids, has not been fully explored; this area remains almost untouched. Only few studies are available describing their utility, majorly in electrochemical applications; however, comprehensive molecular-level insights are still missing. In this work, a detailed physicochemical study of $LiTf_2N$-based DESs consisting of three different HBDs—urea, acetamide, and 2,2,2-trifluoroacetamide—is presented over a wide temperature range. Further knowledge of these DES systems is acquired by changing the mole ratio of $LiTf_2N$:HBD. To illustrate the dipolarity/polarizability and H-bonding strength among the DES systems, photophysical investigation is carried out using solvatochromic probes. This investigation not only draws some parallels among already existing DESs but also opens new avenues in the field of novel solvents by providing crucial information on their nature and understanding their impact on sustainable development.

EXPERIMENTAL METHODS

Materials. All DES constituents, including $LiTf_2N$, acetamide, urea, and 2,2,2-trifluoroacetamide, were acquired from Sigma-Aldrich and used without further purification.

Solvatochromic probes, including 2,6-dichloro-4-(2,4,6-triphenylpyridinium-1-yl)phenolate (betaine dye 33), 4-nitroaniline (NA), and *N*,*N*-diethyl-4-nitroaniline (DENA), were purchased from Fluka, Spectrochem Co. Ltd., and Frinton Laboratories, respectively. Each constituent was stored in a desiccator to avoid contact with moisture. Structures of the constituents used to prepare DESs are presented in Figure 1. A detailed description of the chemicals used is provided in Table 1.

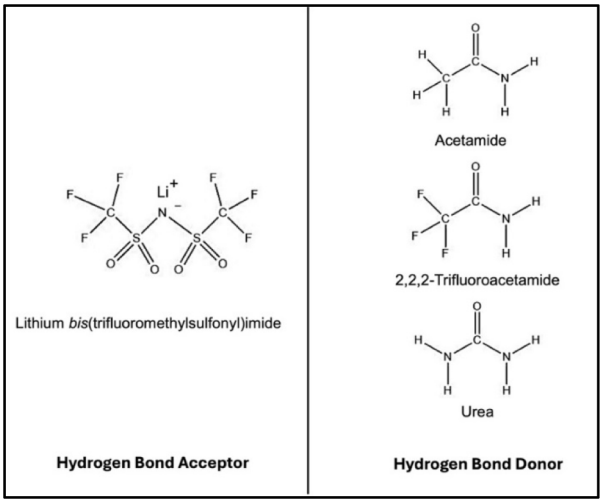


Figure 1. Structures of the components used to prepare the DESs.

Methods. DES of a specific mole ratio was prepared from a precalculated amount of salt, $LiTf_2N$, and the respective HBD using an analytical balance of precision ±0.1 mg, by constant stirring and heating at the same time at about 80 °C for a minimum of 2 h. When a homogeneous mixture was obtained, it was then subjected to vacuum-drying until water was completely ejected. The final water content of DES prior to data acquisition was determined using Karl Fischer titration (KFT) and was found to be less than 100 ppm. Density (ρ) measurements were carried out using a Mettler Toledo DE45 delta range density meter. Measurements were performed in the temperature range 288−358 K. The standard deviation associated with the density measurement is 0.005 g·cm^{-3}. An automated Anton Paar microviscometer (AMVn) with a thermoelectric temperature controller (Peltier-based, with a resolution of 0.01 K and accuracy <0.05 K) was used to study the temperature dependency of dynamic viscosity (η). For improved reproducibility and repeatability of the dynamic viscosity data, the instrument is additionally equipped with various calibrated glass capillaries with sizes of 1.6, 1.8, 3.0, and 4.0 mm. The calculation is based on the classic rolling-ball principle. The deviation in η was estimated to be 2% in the viscosity range spanning between 0.3 and 2500 mPa·s. Electrical conductivity (σ) measurements were carried out on a CM-183 μp-based EC-TDS analyzer with an ATC probe and conductivity cell (CC-03B), purchased from Elico Ltd., India. The standard uncertainty in σ is 2%. The surface tension (γ) was measured at 298 K using a Wilhelmy plate and a DCAT 11EC tensiometer that were obtained from Data-physics. Before each usage, the plate was thoroughly cleaned with Millipore water and rinsed with ethanol to remove any contaminants that might have adhered to it. The plate was heated on a Bunsen burner until it turned hot red and then allowed to cool to room temperature in order to guarantee the total removal of impurities. The standard uncertainty in γ of the solvents is estimated to be 0.05 mN·m^{-1}. The refractive index (n_D) was measured by using a digital ABBE refractometer (Labtronics, LT-738) with a standard uncertainty of 0.0002. By comparing the consistency of the observed refractive indices of distilled water and acetone at room temperature (298 K) with the values described in the literature, the ABBE refractometer instrument was calibrated.

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Table 1. Description of the Chemicals Used in This Work

Chemicals	Molar Mass/g·mol^{-1}	Source	CAS no.	Purity[a] (Mass Fraction)
$LiTf_2N$	287.09	Sigma-Aldrich	90076−65−6	0.99
Acetamide	59.07	Sigma-Aldrich	60−35−5	≥0.99
Urea	60.06	Sigma-Aldrich	57−13−6	≥0.99
2,2,2-Trifluoroacetamide	113.04	Sigma-Aldrich	354−38−1	0.97
Betaine Dye 33	551.69	Fluka	10081−39−7	0.99
DENA	194.23	Frinton Laboratories	2216−15−1	-
NA	138.12	Spectrochem Co. Ltd.	100−01−6	0.99

[a]Purity values are provided by the supplier.

A stock solution of each of the probes was prepared by dissolving the required amount in ethanol in a precleaned amber glass vial, which was stored at 4 ± 1 °C to retard any photochemical reactions. An appropriate amount of the probe solution from the stock was transferred to a 1 cm path-length quartz cuvette. To reach the required final concentration of the probe, ethanol was evaporated using a mild stream of high-purity N_2 gas. The cuvette was immediately filled with a predetermined quantity of DES, and the solution was mixed thoroughly to dissolve the probe. The final concentrations of DENA, NA, and betaine dye 33 were 20, 20, and 50 μM, respectively. The UV−vis molecular absorbance spectra were obtained using a PerkinElmer Lambda 365 double-beam spectrophotometer with a variable bandwidth. Before data processing, all spectra were appropriately adjusted by deducting the spectral responses from the appropriate blanks. Data analysis was performed by using SigmaPlot v15 software.

While Table 2 compares the density and dynamic viscosity of $LiTf_2N$-based DESs to some of the other DESs reported earlier, Table 3 compares the data for $LiTf_2N$-based DESs obtained by us with those reported by other researchers. Our data and several of those reported in the literature show good agreement; however, different water contents, the presence of contaminants, sample sources, and equipment types used for density and viscosity acquisition could constitute possible causes of the measurement inconsistencies.

Table 2. Comparison of Density (ρ/g·cm^{-3}) and Dynamic Viscosity (η/mPa·s) of $LiTf_2N$-Based DESs Studied in This Work with Other Common DESs Reported in the Literature at Pressure p= 0.1 MPa and Temperature T= 298 K[a]

DES[b]	ρ/g·cm^{-3}	η/mPa·s
$LiTf_2N$:acetamide		
1:2	1.540	1185
1:3	1.454	386.6
1:4	1.390	169.8
$LiTf_2N$:2,2,2-trifluoroacetamide		
1:2	1.736	717.2
1:3	1.712	222.9
1:4	1.695	167.3
$LiTf_2N$:urea (1:4)	1.568	1094
ChCl:urea (1:2)[c]	1.212	750
ChCl:glycerol (1:2)	1.1920[d]	259[c]
ChCl:ethylene glycol (1:2)	1.11646[e]	37[c]
ChCl:malonic acid (1:1)	1.231[f]	1124[c]
ChCl:Propionic acid (1:2)[g]	1.08	55.909
La:urea (1:3)[h]	1.804	217.69
Ce:urea (1:3.5)[h]	1.777	173.24
Gd:urea (1:3)[h]	1.877	341.45

[a]Standard Uncertainty u are : $u(\rho)$ = 0.005 g·cm^{-3}, $u_r(\eta)$ = 2%, u(p) = 0.005 MPa, $u(T)$ = 1 K, ChCl: Choline Chloride, La: Lanthanum Nitrate Hexahydrate, Ce: Cerium Nitrate Hexahydrate, Gd: Gadolinium Nitrate Hexahydrate. [b]Data are provided in terms of constituents' mole ratios m:n; Standard Uncertainty with m and n, respectively: $u(m$ or $n) < \pm 0.01$. [c]ref[26]. [d]ref[32]. [e]ref[33]. [f]ref[34]. [g]ref[35]. [h]ref[36].

RESULTS AND DISCUSSION

Temperature and Composition Dependence of Density. Density of $LiTf_2N$-containing DESs with three HBDs—acetamide, urea, and 2,2,2-trifluoroacetamide—in the $LiTf_2N$/HBD mole ratios 1:2, 1:3, and 1:4 as a function of temperature in the range 288 to 358 K is experimentally determined and reported in Table 4. Irrespective of the composition and temperature, the density is found to be the highest for DES with 2,2,2-trifluoroacetamide as the constituent, followed by urea, and the lowest for acetamide. For mole ratio 1:2 and T = 298 K, densities of $LiTf_2N$/2,2,2-trifluoroacetamide, $LiTf_2N$/urea, and $LiTf_2N$/acetamide DESs have the values 1.736 g·cm^{-3}, 1.676 g·cm^{-3}, and 1.540 g·cm^{-3}, respectively. It is important to mention here that the density of a DES is primarily controlled by the intermolecular interactions between the constituents,[27−29] which appear to be the highest in the case of the $LiTf_2N$/2,2,2-trifluoroacetamide DES system. Interestingly, the density of 2,2,2-trifluoroacetamide (0.960 g·cm^{-3} at 298 K) is lower than that of acetamide (1.16 g·cm^{-3} at 298 K) as well as that of urea (1.32 g·cm^{-3} at 298 K). However, 2,2,2-trifluoroacetamide has a much higher molar mass compared to acetamide and urea (113.09 g·mol^{-1} versus 60.06 and 59.07 g·mol^{-1}, respectively), and due to the presence of highly electronegative fluorine atoms in 2,2,2-trifluoroacetamide, the H-bonding and other attractive interactions are the strongest in $LiTf_2N$/2,2,2-trifluoroacetamide DESs as opposed to the other two. This renders the relative volume of the $LiTf_2N$/2,2,2-trifluoroacetamide DESs to become not as high as expected, thus leading to higher associated densities. Fluorine-containing systems are known to exhibit unusual properties.[30,31]

It is noteworthy that type-III $LiTf_2N$-containing DESs are comparatively denser than type-III choline chloride (ChCl)-based DESs (Table 2).[26−35] At ambient conditions, the density of ChCl:HBD (1:2) DESs lies in the range of ∼1.1 g·cm^{-3} to ∼1.2 g·cm^{-3}, which is significantly lower than $LiTf_2N$/HBD DESs. However, type-IV DESs, such as, lanthanum nitrate hexahydrate:urea DESs (La:urea, 1.804 g·cm^{-3} for 1:3), cerium nitrate hexahydrate (Ce:urea, 1.777 g·cm^{-3} for 1:3.5), and gadolinium nitrate hexahydrate (Gd:urea, 1.877 g·cm^{-3} for 1:3) have densities higher than $LiTf_2N$/HBD DESs investigated here.[36] It is convenient to note that the density values of $LiTf_2N$-containing DESs fill the gap between the above two

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Table 3. Comparison of Density (ρ/g·cm^{-3}), Dynamic Viscosity (η/mPa·s), Electrical Conductivity (σ/mS·cm^{-1}), and Surface Tension (γ/mN·m^{-1}) of $LiTf_2N$-Based DESs Obtained in This Work with Those Reported in the Literature at Pressure p = 0.1 MPa and at Specified Temperatures[a]

DES[b]	ρ/g·cm^{-3}		η/mPa·s		σ/mS·cm^{-1}		γ/mN·m^{-1}	
	This Work	Reported	This Work	Reported	This Work	Reported	This Work	Reported
T = 298 K								
$LiTf_2N$:acetamide[c]								
1:2	1.540	-	1185	-	0.337	0.16	33.53	-
1:3	1.454	-	386.6	-	0.767	0.62	37.11	-
1:4	1.390	1.401	169.8	99.56	1.29	1.07	38.07	46.8
T = 303 K								
$LiTf_2N$:2,2,2-trifluoroacetamide[d]								
1:2	1.731	1.650	473.9	134.0				
1:3	1.706	1.601	157.3	120.1				
1:4	1.689	1.539	119.4	59.2				

[a]Standard Uncertainty u are: $u(p)$ = 0.005 MPa, $u(T)$ = 1 K. $u(\rho)$ = 0.005 g·cm^{-3}, $u_r(\sigma)$ = 2%, $u_r(\eta)$ = 2%, $u(\gamma)$ = 0.05 mN·m^{-1}. [b]Data are provided in terms of constituents' mole ratios m:n; Standard Uncertainty with m and n, respectively: $u(m$ or $n) < \pm 0.01$. [c]ref[18]. [d]ref20.

Table 4. Density (ρ/g·cm^{-3}) of $LiTf_2N$-Based DESs at Pressure p = 0.1 and Temperature Range (288 to 358) K[a]

	T/K														
DES[b]	288	293	298	303	308	313	318	323	328	333	338	343	348	353	358
$LiTf_2N$:acetamide															
1:2	1.550	1.544	1.540	1.535	1.528	1.522	1.517	1.511	1.508	1.504	1.500	1.496	1.491	1.486	1.481
1:3	1.464	1.459	1.454	1.450	1.443	1.437	1.433	1.427	1.424	1.420	1.416	1.412	1.408	1.403	1.398
1:4	1.400	1.395	1.390	1.386	1.379	1.374	1.370	1.365	1.361	1.358	1.354	1.350	1.346	1.342	1.337
$LiTf_2N$:2,2,2-trifluoroacetamide															
1:2	1.751	1.744	1.736	1.731	1.721	1.714	1.707	1.699	1.695	1.691	1.686	1.682	1.676	1.671	1.666
1:3	1.727	1.719	1.712	1.706	1.696	1.689	1.682	1.674	1.670	1.666	1.662	1.657	1.652	1.647	1.641
1:4	1.711	1.702	1.695	1.689	1.679	1.672	1.665	1.657	1.653	1.649	1.645	1.640	1.635	1.630	1.624
$LiTf_2N$:urea															
1:2	1.686	1.681	1.676	1.673	1.666	1.660	1.656	1.650	1.646	1.642	1.638	1.633	1.628	1.623	1.618
1:3	1.623	1.618	1.613	1.610	1.603	1.598	1.593	1.588	1.584	1.580	1.576	1.571	1.566	1.561	1.556
1:4	1.577	1.572	1.568	1.565	1.558	1.553	1.549	1.543	1.540	1.536	1.532	1.527	1.523	1.518	1.513

[a]Standard Uncertainty u are: $u(\rho)$ = 0.005 g·cm^{-3}, $u(p)$ = 0.005 MPa, $u(T)$ = 1 K. [b]Data are provided in terms of constituents' mole ratios m:n; Standard Uncertainty with m and n, respectively: $u(m$ or $n) < \pm 0.01$.

types of DESs, offering a broad range of density values across various types of DESs in general.

As expected, irrespective of the composition, density of all three DES systems decreases with increasing temperature; density versus temperature (Figure 2) exhibits excellent linear correlation:

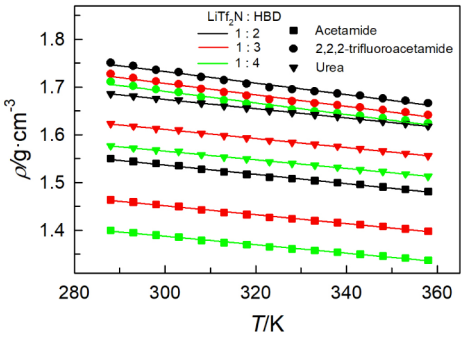


Figure 2. Variation of the density with temperature for the investigated DESs. The solid lines represent fit to the equation $\rho/(\text{g·cm}^{-3}) = \rho_0/(\text{g·cm}^{-3}) + aT/(\text{K})$. Parameters ρ_0 and a, along with R^2 are reported in Table 5.

$$\rho = \rho_0 + aT \quad (1)$$

where ρ/g·cm^{-3} is the density of the DES at temperature T/K, and ρ_0/g·cm^{-3} is the extrapolated density at 0 K, and the extent of change with temperature is expressed as a/g·cm^{-3}·K^{-1}. Table 5 summarizes the recovered parameters obtained, along with their standard deviation for each DES system. Importantly, the a values for $LiTf_2N$/2,2,2-trifluoroacetamide DES at each composition exceed those of $LiTf_2N$/acetamide and $LiTf_2N$/urea DESs (a are similar for the latter two DESs), indicating the volume expansion due to weakening of H-bonds with increase in temperature is highest for 2,2,2-trifluoroacetamide as the HBD. This can again be attributed to the high electronegativity of fluorine atoms which, as the temperature is increased, render the H-bonding relatively weaker in $LiTf_2N$/2,2,2-trifluoroacetamide compared to the other two DES systems.

A change in the ratio of the constituents can bring drastic changes in the physical properties of DESs.[29,36] From Table 4, it is clear that the density of DESs decreases as the concentration of HBD is increased at a given temperature (Figure 3 demonstrates the density trend with the mole ratio). Density appears to decrease linearly with increasing moles of HBD per mole of $LiTf_2N$, regardless of the identity of the HBD; however, the decrease is maximum in the case of

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Table 5. Result of the Regression Analysis of Density (ρ/g·cm^{-3}) versus Temperature (T/K) Data According to Equation: $\rho/(g{\cdot}cm^{-3}) = \rho_{0o}/(g{\cdot}cm^{-3}) + a(T/K)$, for the Investigated $LiTf_2N$-Based DESs Over the Temperature Range 288 to 358 K[a]

DES[b]	ρ_0/g·cm^{-3}	(−) a/g·cm^{-3}·K^{-1} × 10^3	R^2
$LiTf_2N$:acetamide			
1:2	1.96(0.004)	1.00(0.01)	0.998
1:3	1.90(0.003)	0.900(0.009)	0.998
1:4	1.84(0.003)	0.900(0.011)	0.998
$LiTf_2N$:2,2,2-trifluoroacetamide			
1:2	2.09(0.012)	1.20(0.04)	0.987
1:3	2.07(0.013)	1.20(0.04)	0.985
1:4	2.05(0.013)	1.20(0.04)	0.985
$LiTf_2N$:urea			
1:2	1.83(0.006)	1.00(0.019)	0.995
1:3	1.73(0.006)	0.900(0.018)	0.995
1:4	1.65(0.005)	0.900(0.017)	0.995

[a]Standard Uncertainty u are $u(\rho)$ = 0.005 g·cm^{-3}, $u(p)$ = 0.005 MPa, $u(T)$ = 1 K. Standard deviations are given within parentheses. [b]Data are provided in terms of constituents' mole ratios m:n; Standard Uncertainty with m and n, respectively: $u(m$ or $n) < \pm 0.01$.

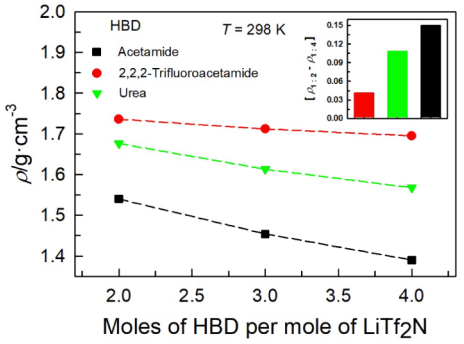


Figure 3. Densities of $LiTf_2N$-based DESs as a function of HBD per mole of $LiTf_2N$ at 298 K. Inset represents the degree of change evaluated as the difference between density of mole ratios 1:2 and 1:4.

acetamide and minimum for 2,2,2-trifluoroacetamide as HBDs. This contrast with the temperature sensitivity of the density. The decrease in density as concentration of HBD is increased is easy to comprehend, as $LiTf_2N$ has a significantly higher density (2.15 g·cm^{-3} at 293 K) compared to those of HBDs used in our investigation (vide supra).

Change in the molar ratio accompanies the change in molar mass (M/g·mol^{-1}) of a DES system, which is determined for different compositions according to the following relation:

$$M = n_1M_1 + n_2M_2 \quad (2)$$

where M_1 and M_2 are the molar masses of the two components, and n_1 and n_2 represent their respective number of moles.[37] As already mentioned, the physical properties of DESs can be altered by altering the composition of their constituents. A handful of properties can be conveniently obtained with knowledge of their density values. Some are empirically determined at 298 K and are listed in Table 6, which includes molar volume (V/cm^3·mol^{-1}), molecular volume (V_m/nm^3), standard molar entropy (S_0/J·K^{-1}), and lattice energy (U_{POT}/kJ·mol^{-1}).[36] The following equations are used for their estimation:

Table 6. Summary of Estimated Physicochemical Property Values Associated with the Density for $LiTf_2N$-based DESs at a Pressure p = 0.1 MPa and a Temperature T = 298 K[a]

DES[b]	M/g·mol^{-1}	V/cm^3·mol^{-1}	V_m/nm^3	S_0/J·K^{-1}	U_{POT}/kJ·mol^{-1}
$LiTf_2N$:acetamide					
1:2	405.2	263.1	0.4369	574.1	413.0
1:3	464.3	319.3	0.5302	690.4	393.7
1:4	523.4	376.5	0.6252	808.8	378.2
$LiTf_2N$:2,2,2-trifluoroacetamide					
1:2	513.2	295.6	0.4909	641.4	401.2
1:3	626.2	365.8	0.6074	786.6	380.8
1:4	739.2	436.1	0.7242	932.2	365.0
$LiTf_2N$:urea					
1:2	407.2	242.9	0.4034	532.3	421.3
1:3	467.3	289.7	0.4810	629.1	403.2
1:4	527.3	336.3	0.5584	725.5	388.7

[a]Standard Uncertainty u are $u(p)$ = 0.005 MPa, $u(T)$ = 1 K. [b]Data are provided in terms of constituents' mole ratios m:n; Standard Uncertainty with m and n, respectively: $u(m$ or $n) < \pm 0.01$.

$$V = \frac{M}{\rho} \quad (3)$$

$$V_m = \frac{M}{\rho N_A} \quad (4)$$

$$S_0 = 1246.5 + 29.5 \quad (5)$$

$$U_{POT} = 1981.2\left(\frac{\rho}{M}\right)^{1/3} + 103.8 \quad (6)$$

In comparison, the $LiTf_2N$/urea DES has a lower molar volume at each ratio, whereas the $LiTf_2N$/2,2,2-trifluoroacetamide DES exhibits a higher molar volume (242.9 cm^3·mol^{-1} versus 295.6 cm^3·mol^{-1} for 1:2, respectively). Phung et al. have proposed a ring-type structure of the $LiTf_2N$/2,2,2-trifluoroacetamide DES, where Li^+ coordinates with the CF_3 group as well with the CO of the 2,2,2-trifluoroacetamide molecule.[20] We believe that this molecular arrangement could lead to the occupation of a larger volume. The strength of intermolecular interactions is again manifested in the values of standard molar entropy and lattice energy, where we found that the lattice energy of $LiTf_2N$/urea DES is higher, whereas that of $LiTf_2N$/2,2,2-trifluoroacetamide DES is lower. The opposite order is justified for the standard molar entropy.

So far, only a few reports are available for the physicochemical characterization of $LiTf_2N$-based DESs. Some are compared in Table 3 with the data obtained in this study. Chen and coworkers have reported a density of 1.401 g·cm^{-3} for the $LiTf_2N$/acetamide DES in a 1:4 mol ratio, which is fairly close to this work.[18] The density reported for $LiTf_2N$/2,2,2-trifluoroacetamide by Phung et al. differs to some extent from this work, possibly due to the highly volatile nature of 2,2,2-trifluoroacetamide, which makes itrelatively difficult to use during DES preparation and data acquisition.[20]

Dynamic Viscosity and Its Temperature and Composition Dependence. Table 7 lists the measured dynamic viscosity (η/mPa·s) of $LiTf_2N$-based DESs in the temperature range 288 to 358 K. Dynamic viscosity of a solvent controls the kinetics of many inter- as well as intramolecular processes; therefore, it is intuitive to describe it as a function of DES constituents and temperature.[28,38,39] For a given mole ratio and temperature, dynamic viscosity decreases as: $LiTf_2N$/urea >

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Table 7. Dynamic Viscosity (η/mPa·s) of $LiTf_2N$-Based DESs at Different Mole Ratios at Pressure p = 0.1 MPa and Temperature T = (288 to 358) K[a]

	T/K												
DES	288	293	298	303	308	313	318	323	328	333	338	343	348	353	358
$LiTf_2N$:acetamide															
1:2	n.d.	1895	1185	805.9	554.5	392.6	285.1	212.6	157.5	122.9	97.24	78.48	64.39	54.01	45.55
1:3	853.0	559.3	386.6	266.8	195.0	144.5	108.4	85.06	66.59	53.87	44.47	36.98	31.36	27.26	23.98
1:4	357.8	241.5	169.8	123.1	91.76	69.96	54.37	43.21	34.83	28.54	23.60	19.82	16.80	14.40	12.64
$LiTf_2N$:2,2,2-trifluoroacetamide															
1:2	n.d.	1138	717.2	473.9	326.0	230.3	169.3	127.1	100.6	79.24	63.10	49.75	40.92	33.81	30.82
1:3	559.2	342.5	222.9	157.3	105.2	79.60	61.22	48.38	38.72	31.53	26.02	21.91	18.55	15.89	13.76
1:4	365.3	242.1	167.3	119.4	87.93	66.34	51.39	40.57	32.62	26.65	22.12	18.54	15.73	13.48	11.65
$LiTf_2N$:urea															
1:2	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	1815	1208	831.5	581.9	428.5	320.6	244.5	189.3	150.2
1:3	n.d.	n.d.	n.d.	1519	967.8	643.8	445.4	321.1	237.8	180.7	140.2	109.9	88.26	72.11	59.72
1:4	n.d.	1824	1094	698.3	467.1	325.1	233.9	174.6	132.7	102.8	81.27	65.53	53.55	44.50	37.66

[a]n.d. (not determined): Due to the limitation of the technique, data for more than 2500 mPa s viscosity could not be acquired. Standard Uncertainty u are: $u_r(\eta)$ = 2 %, $u(p)$ = 0.005 MPa, $u(T)$ = 1 K.

$LiTf_2N$/acetamide > $LiTf_2N$/2,2,2-trifluoroacetamide. It is usually inferred that the knowledge of the dynamic viscosity of a medium can be related to the strength of intermolecular interactions present among the constituents, which, in the case of DESs, are mainly H-bonding, among others.[40,41] In general, the stronger the interactions, the higher the viscous drag. As already seen in Table 6, the trends in standard molar entropy and lattice energy follow the trend observed in dynamic viscosity among the three DES systems. Interestingly, the viscosity of the $LiTf_2N$/urea DES is much higher than that of the urea-based type-III DES with choline chloride in a 1:2 (ChCl:urea) mole ratio (750 mPa·s at 298 K)[26] and the type-IV DES with La and Gd in a 1:3 (La/Gd:urea) mole ratio (217.69/341.45 mPa·s at 298 K)[36] (Table 3).

As the temperature increases, molecules gain thermal energy and start vibrating at their mean position rapidly. This leads to disruption of the H-bonds and consequently lowers the dynamic viscosity. To analyze the temperature dependence, ln η is plotted against $1/T$ for all DES systems (Figure 4A) and fitted according to the Vogel–Fulcher–Tammann (VFT) equation, with goodness-of-fit in terms of R^2 reported in Table 8:

$$\ln \eta = A + \frac{B}{T - T_o} \tag{7}$$

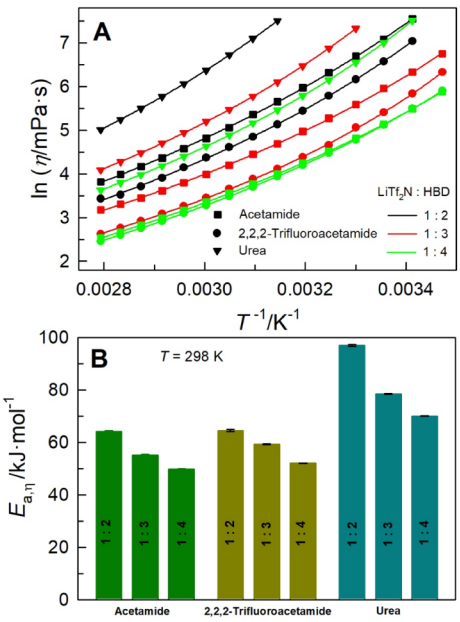


Figure 4. Variation of dynamic viscosity of $LiTf_2N$-based DESs as a function of temperature (panel A). Solid lines represent the best fit to the VFT model ln η/(mPa·s) = $A + \frac{B}{T - T_o}$. Panel B shows a comparison of the activation energy of the viscous flow ($E_{a,\eta}$) at 298 K for these DESs. The recovered parameters, along with R^2, are reported in Table 8.

where A and B B = are empirical, material-dependent parameters, and T_0 is also an empirical fitting parameter, which typically lies about 50 °C below the glass transition temperature[42] (the recovered parameters are reported in Table 8). Activation energies of viscous flow ($E_{a,\eta}$) were estimated at 298 K using:

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Table 8. Summary of Parameters Associated with Dynamic Viscosity of DESs According to the VFT Model: $\mathrm{Ln}\eta/(\mathrm{mPa\cdot s}) = A + \frac{B}{T - T_o}$

DES[a]	(−) A	B	T_0/K	R^2	[b]$E_{a,\eta}$/kJ·mol^{-1}
$LiTf_2N$:acetamide					
1:2	1.91(0.18)	935(47)	194(3)	0.999	64.1(0.4)
1:3	1.43(0.15)	726(36)	199(3)	0.999	55.2(0.3)
1:4	2.19(0.04)	795(10)	189(1)	0.999	49.8(0.1)
$LiTf_2N$:2,2,2-trifluoroacetamide					
1:2	1.55(0.21)	755(49)	205(4)	0.999	64.5(0.4)
1:3	1.12(0.10)	530(20)	217(2)	0.999	59.4(0.2)
1:4	2.05(0.01)	729(3)	196(1)	0.999	52.1(0.1)
$LiTf_2N$:urea					
1:2	1.53(0.16)	949(40)	213(3)	0.999	97.0(0.3)
1:3	1.18(0.05)	762(13)	213(1)	0.999	78.5(0.1)
1:4	1.28(0.03)	723(8)	210(1)	0.999	70.0(0.1)

[a]Data are provided in terms of constituents' mole ratios m:n; Standard Uncertainty with m and n, respectively: $u(m \text{ or } n) < \pm 0.01$. [b]$E_{a,\eta}$ is calculated at T = 298 K. Standard deviations are given within parenthesis.

$$E_{a,\eta} = RB\left(\frac{T}{T - T_o}\right)^2 \quad (8)$$

and are compared in Figure 4B. $E_{a,\eta}$ for $LiTf_2N$/urea DES (the DES system with the highest dynamic viscosity) is significantly higher at each composition (97.0 ± 0.3 kJ·mol^{-1} for 1:2) in comparison to the $LiTf_2N$/acetamide and the $LiTf_2N$/2,2,2-trifluoroacetamide DES systems (they have similar $E_{a,\eta}$; 64.1 ± 0.4 kJ·mol^{-1} and 64.5 ± 0.4 kJ·mol^{-1}, respectively, for 1:2). The higher energy required for molecular species to be set in motion against frictional forces is again highlighted for the $LiTf_2N$/urea DES system. The $E_{a,\eta}$ values were found to decrease as the HBD concentration is increased in each $LiTf_2N$/HBD DES system. This indicates that the addition of HBD reduces the hindrance to viscous flow due to a decrease H-bonding interaction. The $E_{a,\eta}$ for ChCl:urea (1:2), ChCl:glycerol (1:2), ChCl:malonic acid (1:1), and ChCl:oxalic acid (1:1) are reported to be 55.7[31]/51.2[43] kJ·mol^{-1}, 45.1[44]/42.3[43] kJ·mol^{-1}, 46.66[34]/44.6[45] kJ·mol^{-1}, 65.20[34]/59.2[45] kJ·mol^{-1}, respectively. In comparison, for ionic liquids (ILs), $[C_4C_1im][PF_6]$, $[C_4C_1im][CH_3CH_2OSO_3]$, $[C_4C_1im][CF_3SO_3]$, $[C_4C_1im][(CF_3SO_2)_2N]$, and $[C_2C_1im][(CF_3SO_2)_2N]$, the $E_{a,\eta}$ are 47.19 kJ·mol^{-1}, 44.13 kJ·mol^{-1}, 34.48 kJ·mol^{-1}, 30.14 kJ·mol^{-1}, and 25.59 kJ·mol^{-1}, respectively.[46] It should be noted that the $E_{a,\eta}$of the ChCl-based DESs were calculated using the Arrhenius relation, while for ILs, the VFT equation was used. Thus, although the $E_{a,\eta}$ values for $LiTf_2N$-containing DESs are closer to those for type-III DESs, their viscosity−temperature dependence is more similar to that of ILs.

Apart from the temperature effects, the composition-driven changes in dynamic viscosity are also important and are presented in Figure 5. It is observed that at a given temperature with increasing HBD concentration, dynamic viscosity decreases monotonically. However, the effect appears to be more profound at lower temperatures compared to that at higher temperatures, where viscosity values become almost similar. It is apparent, however, that excess HBD may lead to decrease in H-bonding interactions, combined with a possible increase in electrostatic interaction between Li^+ and Tf_2N^-.[20]

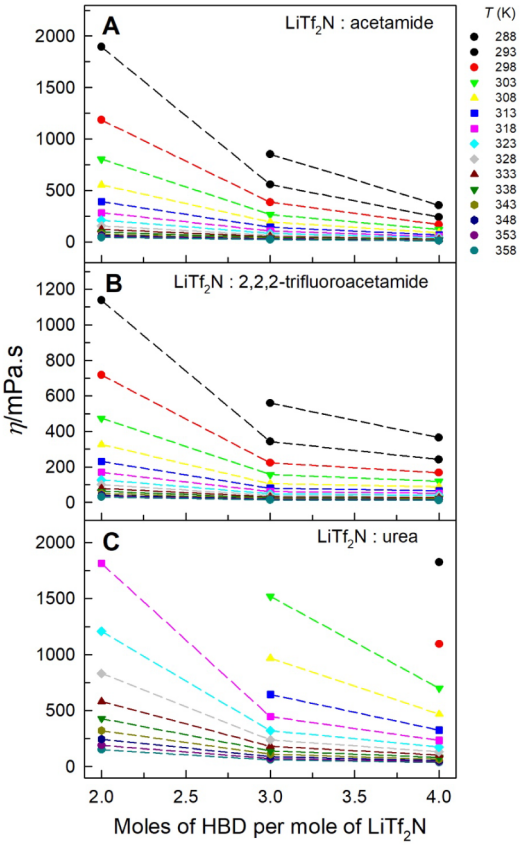


Figure 5. Variation of dynamic viscosity with moles of HBD per mole of $LiTf_2N$ for the DESs $LiTf_2N$:acetamide, $LiTf_2N$:2,2,2-trifluoroacetamide, and $LiTf_2N$:urea (panels A, B, and C, respectively) in the temperature range of 288 to 358 K.

Electrical Conductivity, Surface Tension, and Refractive Index. Electrical conductivity (σ), surface tension (γ), and refractive index (n_D) of $LiTf_2N$-containing DESs at 298 K are measured experimentally and reported in Table 9. The information is crucial, as it may afford valuable insight into the

Table 9. Electrical Conductivity (σ/mS·cm^{-1}), Surface Tension (γ/mN·m^{-1}), and Refractive Index (n_D) of $LiTf_2N$-Based DESs at Pressure p = 0.1 MPa and Temperature T = 298 K[a]

DES[b]	σ/mS·cm^{-1}	γ/mN·m^{-1}	n_D
$LiTf_2N$:acetamide			
1:2	0.337	33.53	1.3941
1:3	0.767	37.11	1.4055
1:4	1.29	38.07	1.4135
$LiTf_2N$:2,2,2-trifluoroacetamide			
1:2	0.309	21.97	1.3639
1:3	0.809	28.00	1.3602
1:4	1.30	27.55	1.3575
$LiTf_2N$:urea			
1:2	0.0900	31.09	1.4186
1:3	0.219	40.95	1.4330
1:4	0.341	42.50	1.4420

[a]Standard Uncertainty u are $u(p)$ = 0.005 MPa, $u(T)$ = 1 K, $u_r(\sigma)$ = 2%, $u(\gamma)$ = 0.05 mN·m^{-1}, $u(n_D)$ = 0.0002. [b]Data are provided in terms of constituents' mole ratios m:n; Standard Uncertainty with m and n, respectively: $u(m \text{ or } n) < \pm 0.01$.

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packing of molecules via inter-/intramolecular interactions among constituents within the DES systems.[47,48] Moreover, these properties are highly dependent on the constituents of eutectic mixtures as well as their composition. The electrical conductivity of DESs is one of the key properties to elucidate the effectiveness of the electrolyte in Li-ion batteries.[24,49] It is found that the electrical conductivity of $LiTf_2N$/urea DESs is significantly lower than those of $LiTf_2N$/acetamide and $LiTf_2N$/2,2,2-trifluoroacetamide DESs; the latter two have similar electrical conductivity at 298 K. The monoamide family HBDs (acetamide and 2,2,2-trifluoroacetamide)- containing DESs appear to possess higher electrical conductivity in comparison to urea, a diamide HBD-containing DESs. For similar components, in general, electrical conductivity may vary inversely with solvent dynamic viscosity.[20,50] Therefore, it is reasonable to expect that $LiTf_2N$/urea DESs exhibit the lowest electrical conductivity in comparison to the other two DESs. Furthermore, when the constituents exhibit a higher capability of coordinating intermolecularly, the free volume required for the facile movement of ions decreases, resulting in a decrease in electrical conductivity.[20] Similar electrical conductivities of $LiTf_2N$/acetamide and $LiTf_2N$/2,2,2-trifluoroacetamide DESs, despite the fact that dynamic viscosity is lower for the 2,2,2-trifluoroacetamide-containing DESs, on the other hand, hint toward the structural differences (H-bonding and other interactions) of these two DES systems. Irrespective of the type of HBD, electrical conductivity is found to increase monotonically with an increase in the relative amount of HBD within the DES system (Figure 6A). Two factors are primarily responsible for the rise in electrical conductivity upon an increase in HBD concentration. First, as already discussed, an increase in the concentration of HBD lowers the dynamic viscosity of the DES, resulting in an increase in electrical conductivity. Second, higher concentration of HBD promotes the dissociation of $LiTf_2N$ into ions. It is reported that HBD enables coordination with both Li^+ and Tf_2N^- and acts as a complexing agent; in general, Li^+ solvates 4−6 solvent molecules.[20] It is evident that the urea-containing DES is least sensitive to the mole ratio variation in comparison to acetamide- and 2,2,2-trifluoroacetamide-containing DESs. Nevertheless, this indicates that urea, as the HBD, is unable to produce a greater impact on either the reduction in dynamic viscosity or the degree of dissociation of $LiTf_2N$.

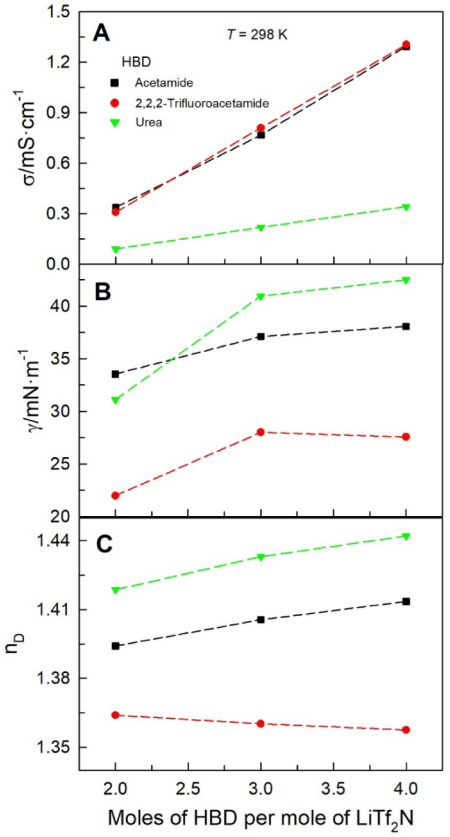


Figure 6. Electrical conductivity (panel A), surface tension (panel B), and refractive index (panel C) of the investigated DESs as a function of $LiTf_2N$:HBD mole ratio at 298 K.

Surface tension is a useful property to comprehend the strength of cohesive molecular interactions within a liquid.[51,52] For DES systems, apart from temperature, surface tension depends on the constituents and their affinities toward each other.[52] Overall, it is a measure of the cohesive forces within a liquid milieu. Measured surface tension of the DES systems at 298 K (Table 9) reveal that the $LiTf_2N$/2,2,2-trifluoroacetamide DES system exhibits the lowest surface tensions among the three systems investigated. Tentatively, it is attributed to the presence of electronegative fluorine atoms in 2,2,2-trifluoroacetamide. The surface tension of the $LiTf_2N$/urea DES is found to be higher than its acetamide counterpart for the mole ratios 1:3 (40.95 versus 37.11 $mN{\cdot}m^{-1}$) and 1:4 (42.50 versus 38.07 $mN{\cdot}m^{-1}$), but lower for the molar ratio 1:2 (31.09 versus 33.53 $mN{\cdot}m^{-1}$). Interestingly, increasing the concentration of HBD leads to an increase in the surface tension for all three DES systems (Figure 6B). They exhibit similar variation in surface tension where it rises significantly when the $LiTf_2N$/HBD mole ratio changes from 1:2 to 1:3 but an insignificant change is noticed when the ratio is further increased to 1:4. Cohesiveness within DES systems appears to increase as concentration of $LiTf_2N$ is decreased (i.e., HBD is increased).

The refractive index is an important property as far as fabrication of optical devices is concerned.[53,54] Measured refractive index values of $LiTf_2N$-containing DES systems at 298 K are listed in Table 9. At a given composition, the refractive index is found to follow the trend: $LiTf_2N$/urea > $LiTf_2N$/acetamide > $LiTf_2N$/2,2,2-trifluoroacetamide. The observed trend follows the order of the H-bonding capacity of the HBDs. Interestingly, the refractive index increases with an increase in HBD concentration when acetamide and urea are the HBDs, and it decreases for the $LiTf_2N$/2,2,2-trifluoroacetamide DES system (Figure 6C). This again highlights the distinction in the context of fluorous compounds and their capabilities in forming intermolecular interactions within such systems.[30] Zhang and coworkers have shown that the refractive index of a DES can be tuned by changing one component or changing its composition, and they have explicitly demonstrated that the nature of HBA and HBD plays a vital role. Their work significantly aligns with the results obtained in this work, where the refractive index is found to be sensitive to both the identity of the HBD as well as the composition of the DES.[54]

E_T^N and Kamlet−Taft Parameters. It is well-established that 2,6-diphenyl-4-(2,4,6-triphenylpyridinium-1-yl)phenolate (betaine dye 30) exhibits an unusually high solvatochromic band shift. The lowest energy intramolecular charge-transfer

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absorption band of betaine dye 30 is hypsochromically shifted by ca. 357 nm, from the relatively nonpolar diphenyl ether (λ_{max} ~ 810 nm) to the highly polar water (λ_{max} ~ 453 nm).[55,56] It has been determined that the differential solvation of betaine dye 30's highly polar equilibrium ground-state and the less polar first Franck−Condon excited state, with increasing solvent polarity, is the source of its negative solvatochromism.[55,56] The pyridinium portion of the zwitterionic molecule receives significant charge transfer from the phenolate. Due to its zwitterionic nature, the solvent's HBD acidity has a significant impact on the solvatochromic probe behavior of betaine dye 30. Solvents that donate H-bonds stabilize the ground state more than the excited state. The empirical scale of solvent "polarity," $E_T(30)$ for betaine dye 30, is defined as the molar transition energy of the dye, traditionally expressed in kcal·mol^{-1}, according to the expression $E_T(30) = 28591.5/\lambda_{max}$ in nm.[55,56] A derivative of betaine dye 30, 2,6-dichloro-4-(2,4,6-triphenylpyridinium-1-yl)phenolate (betaine dye 33), is utilized in this work to study $LiTf_2N$-based DESs because it has certain advantages over betaine dye 30. For our experiments, betaine dye 30 is not appropriate due to its limited solubility in several H-bonded solvent systems. However, because of its inherent structural differences from betaine dye 30, betaine dye 33 does not have these issues. For historical reasons, it has been related to number 33, and the lowest energy absorbance transition of this dye [i.e., $E_T(33)$] is calculated in the same way as $E_T(30)$ is calculated.[57]

Thus, from the absorbance spectra of betaine dye 33, the corresponding $E_T(33)$ are estimated and converted into E_T^N using eqs 9 and 10:

$$E_T(30) = 0.9953(\pm 0.0287) \times E_T 33 - 8.1132(\pm 1.6546) \quad (9)$$

$R = 0.9926$, standard error of estimate = 0.8320, $n = 20$.

$E_T(30)$ was obtained from $E_T(33)$ (i.e., eq 9) by estimating the lowest energy Uv−Vis absorbance band maxima for both dyes in 20 different solvents and performing linear regression analysis between the two.

$$E_T^N = \frac{[E_T(30)_{Solvent} - E_T(30)_{TMS}]}{[E_T(30)_{Water} - E_T(30)_{TMS}]} \quad (10)$$

Here, TMS stands for tetramethylsilane. From $E_T(30)_{Water}$ = 63.1 kcal·mol^{-1} and $E_T(30)_{TMS}$ = 30.7 kcal·mol^{-1}, we obtain

$$E_T^N = \frac{[E_T(30)_{Solvent} - 30.7]}{32.4} \quad (11)$$

E_T^N is easier to conceive as it is a dimensionless quantity.

The Uv−Vis molecular absorbance spectra of betaine dye 33 dissolved in $LiTf_2N$-containing DESs composed of different HBDs in $LiTf_2N$:HBD (1:2) mole ratios at 298 and 358 K are presented in Figure 7 (absorbance spectra at all temperatures are provided in Figure S1). A careful examination of the spectra reveals that the wavelength maxima of the lowest energy absorbance band of the dye follow the trend: $LiTf_2N$/acetamide < $LiTf_2N$/urea < $LiTf_2N$/2,2,2-trifluoroacetamide. Table 10 lists the estimated E_T^N values for the investigated DESs in the temperature range 298 to 358 K [corresponding $\lambda_{max,33}$ and $E_T(30)$ values are provided in Table S1]. Regardless of the temperature, the highest and the lowest E_T^N values are observed for the DESs composed of acetamide and 2,2,2-trifluoroacetamide, respectively. It is noteworthy that E_T^N values for all DES systems within the investigated temperature range are ≥1.00(±0.01), implying these DESs possess considerably high dipolarity/polarizability and/or HBD acidity, with the highest E_T^N of 1.10 for the $LiTf_2N$/acetamide DES at a 1:2 mol ratio at 298 K. It is clear that E_T^N does not change much by changing the identity of HBD. Recently, DESs formed by mixing urea with hydrated lanthanide salts were reported to possess exceptionally high E_T^N values ranging from 1.29 to 1.42.[58] It is interesting to note that the E_T^N values for type III DESs, such as those formed by mixing different HBDs with ChCl, are found to lie in the range 0.60 to 0.90.[59] The effect of Li-salt on E_T^N values is well documented by Pandey et al., who earlier reported that adding Li-salt to ChCl:glycerol (1:2) can enhance E_T^N values to a significant amount due to favorable interactions of Li^+ with the −OH of glycerol.[60]

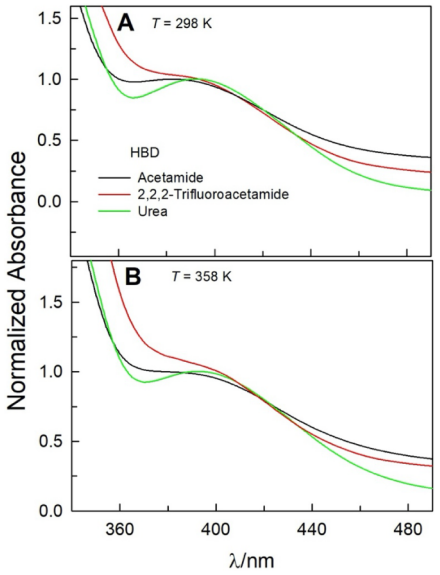


Figure 7. Absorbance spectra of Reichardt's betaine dye 33 (50 μM) in $LiTf_2N$-based DESs in a $LiTf_2N$:HBD mole ratio of 1:2 at 298 K (panel A) and 358 K (panel B).

With an increase in temperature, the E_T^N decreases for each investigated DES system. A decrease in E_T^N implies a decrease in the dipolarity/polarizability and/or the HBD acidity of the medium as the temperature rises. It is often argued that "polarity" tends to diminish as temperature rises because polar liquids undergo greater average thermal reorientation, as a result of the cooperative effect being partially destroyed.[59,61] The change in E_T^N as a function of temperature is presented in Figure 8 (analysis of the linear fit of the data is provided in Table S2). It is reported that E_T^N values of DESs consist of amide-based HBDs are comparably less affected by the temperature variation than those having HBDs of −OH functionalities.[59,61] Pandey et al. have reported E_T^N values for some common ChCl-based DESs and have shown that temperature sensitivity of E_T^N is lowest (-0.7×10^{-3} K^{-1}) when ChCl is mixed with urea among different HBDs (ethylene glycol, glycerol, and urea). Moreover, their sensitivity is found to be similar within ILs [bmim][PF_6] (-0.8×10^{-3} K^{-1}) and [bmim][BF_4] (-0.7×10^{-3} K^{-1}).[62]

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Table 10. Estimated Parameters E_T^N, π^*, α, and β for $LiTf_2N$-Based DESs at Temperature T= (298 to 358) K[a]

T/K	$LiTf_2N$:acetamide (1:2)[b]				$LiTf_2N$:2,2,2-trifluoroacetamide (1:2)[b]				$LiTf_2N$:urea (1:2)[b]			
	E_T^N	π^*	α	β	E_T^N	π^*	α	β	E_T^N	π^*	α	β
298	1.10	1.40	0.96	0.10	1.03	1.57	0.67	0.14	1.06	1.61	0.69	0.14
308	1.09	1.38	0.95	0.10	1.04	1.56	0.69	0.08	1.06	1.59	0.69	0.07
318	1.08	1.36	0.94	0.09	1.03	1.53	0.69	0.09	1.05	1.59	0.69	0.07
328	1.07	1.34	0.95	0.09	1.02	1.53	0.67	0.12	1.05	1.57	0.68	0.07
338	1.07	1.33	0.95	0.08	1.01	1.54	0.65	0.06	1.04	1.56	0.69	0.11
348	1.06	1.33	0.94	0.06	1.00	1.49	0.67	0.07	1.03	1.54	0.69	0.08
358	1.05	1.33	0.93	0.06	0.99	1.48	0.67	0.06	1.03	1.53	0.69	0.07

[a]Standard Uncertainty u are $u(T)$ = 1 K, $u(\lambda)$ = 1 nm. [b]Data are provided in terms of constituents' mole ratios m:n; Standard Uncertainty with m and n, respectively: $u(m \text{ or } n) < \pm 0.01$.

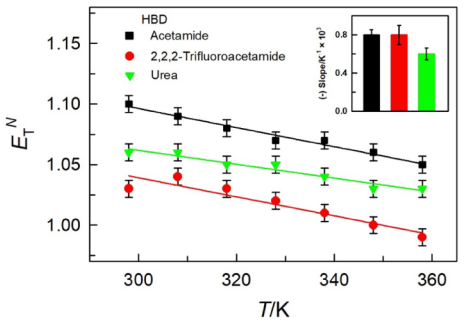


Figure 8. Plots of E_T^N as a function of temperature for $LiTf_2N$:HBD (1:2) DESs. Slopes obtained for each DES system are compared in the inset.

The Kamlet–Taft empirical parameters of solvent polarity——dipolarity/polarizability (π^*), HBD acidity (α), and HBD basicity (β)—were estimated next.[60,63−67] The π^* is estimated from the absorption maximum (ν_{DENA}, in kK) of DENA, a non-H-bond donor solute, using[60,67]

$$\pi^* = 8.649 - 0.314\nu_{DENA} \quad (12)$$

and then α was estimated from $E_T(30)$, estimated earlier, and π^* values.[60,66]

$$\alpha = [E_T(30) - 14.6(\pi^* - 0.23\delta) - 30.31]/16.5 \quad (13)$$

The δ parameter in eq 13 is a "polarizability correction term," equal to 0.0 for nonchlorinated aliphatic solvents, 0.5 for polychlorinated aliphatics, and 1.0 for aromatic solvents.[65] Finally, β values are determined from the enhanced solvatochromic shift of NA relative to its homomorph DENA, $-\Delta\nu$(DENA−NA)/kK:[64]

$$\beta = -0.357\nu_{NA} - 1.176\pi^* + 11.12 \quad (14)$$

Tables 10 and S1 list the estimated π^*, α, and β values and their corresponding absorption wavelengths of dyes DENA and NA, respectively, in the investigated DES systems. It is interesting to note that while $LiTf_2N$/urea exhibits the highest π^*, followed by $LiTf_2N$/2,2,2-trifluoroacetamide and $LiTf_2N$/acetamide (1.61, 1.57, and 1.40, respectively, at 298 K), the α is considerably higher for acetamide DES in comparison to 2,2,2-trifluoroacetamide and urea DESs (0.96 versus 0.67 and 0.69, respectively, at 298 K). The combination of higher π^* and lower α and vice versa results in similar E_T^N values of the investigated DES systems as ~68% of the contribution to E_T^N is from the HBD acidity of the milieu.[56,57] In comparison, DESs formed by mixing urea with hydrated lanthanide salts have been reported to exhibit exceptionally high π^*; a π^* of 1.70 is exhibited by gadolinium salt (Gd)-containing DES (Gd: urea in a 1:2 molar ratio). The π^* ranged from 1.50 to 1.70 for the lanthanide salts-urea DESs.[58] In this context, the $LiTf_2N$-containing DES systems can also be considered to exhibit significantly high dipolarity/polarizability. In comparison, for choline chloride (ChCl)-based DESs, the reported π^* is only 1.23 for (ChCl:urea:: 1:2),[59] 1.373 for (ChCl:citric acid:: 1:1),[68] and 1.406 for (ChCl:$ZnCl_2$:: 1:2).[68] The highest π^* for any DES that is reported earlier was 1.671 for (betaine:citric acid:1:1)[68] under ambient conditions.

The vis-à-vis absorption maxima of both DENA and NA, respectively, show a decrease upon increasing temperature (Figures S2 and S3). While not much variation is observed in HBA basicity of DESs with change in temperature, as expected, the dipolarity/polarizability does decrease monotonically with increasing temperature. The increased average thermal reorientation of the dipoles leads to the destruction of the cooperative effect, resulting in decreased dielectric constant with increasing temperature. While the $LiTf_2N$-containing DESs investigated in this work offer considerably high HBD acidity (α), especially by $LiTf_2N$/acetamide DES (vide supra), the HBA basicity (β) of these DES systems is fairly low (ranging from 0.06 to 0.14 only, with no apparent trend). This, combined with the estimated E_T^N values, shows widely varying physicochemical properties offered by the $LiTf_2N$-containig DES systems investigated here.

CONCLUSIONS

Li-salt-based polar DESs are found to possess widely varying physicochemical properties that depend on the identity of the HBD and the composition. The densities of $LiTf_2N$-containing DESs with urea/acetamide/2,2,2-trifluoroacetamide are in between those of ChCl/HBD and lanthanide nitrate hexahydrate salt/urea DESs with similar mole ratios, thus filling the gap in the density of DESs of various types and classes. Dynamic viscosities also vary considerably depending on the identity of the HBD and the composition of these $LiTf_2N$-containing DESs. Electrical conductivity, surface tension, and refractive index of the $LiTf_2N$-based DESs complement those of common solvents, ionic liquids, and other DESs. These Li-salt-based DESs afford appreciable dipolarity/polarizability and H-bond donating acidity, but offer poor H-bond accepting basicity. Based on the spans of various physicochemical properties reported here, the Li-salt/HBD DESs have potential to replace conventional toxic media in a variety of subdisciplines, including synthesis, separation, and

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extraction, but most notably in electrochemistry and energy storage.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jced.5c00153.

Table S1 lists the absorption wavelength maxima of solvatochromic probes used. Table S2 provides linear regression parameters of plot of E_T^N versus T (K). Figures S1−S3 show absorbance spectra of Reichardt's betaine dye 33, DENA, and NA, respectively, in lithium salt based DESs (PDF)

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Notes

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